

Self-Supervised Learning for Video Geometry

Vanja Kovinic

1 Introduction

Recent advances in 3D vision have produced feed-forward models that reconstruct geometry without optimization. VGGT [1] predicts camera poses, depth, and structure from multi-view images in under one second; Geo4D [2] extends to dynamic scenes using video diffusion priors. Both rely on labeled or synthetic supervision, limiting scalability to billions of hours of unlabeled video online.

Videos offer an extraordinarily rich yet underutilized signal for geometric learning. Unlike static images, videos provide: (1) natural multi-view geometry through camera motion, (2) temporal consistency constraints that implicitly encode 3D structure, (3) motion cues that reveal depth ordering and object boundaries, and (4) virtually unlimited scale - billions of hours of video exist without geometric labels. Self-supervised learning transformed language [3] and vision [4, 5] by exploiting unlabeled data. Can it do the same for video geometry?

Self-supervised monocular depth exists [6], but methods focus on per-frame predictions. Designing objectives that teach how 3D structure evolves over time remains unexplored. Which temporal self-supervised objectives best capture 4D structure? What geometric understanding emerges from unlabeled videos?

2 Background

2.1 Feed-Forward Multi-View Reconstruction

Multi-view reconstruction traditionally requires optimization-based structure-from-motion [7]. Recent work replaces this with feed-forward prediction. DUST3R [8] pioneered dense point regression from image pairs but requires global alignment for multiple views. VGGT [1] extends this with camera tokens and alternating attention, processing 1-200+ images jointly. MonST3R [9] performs pairwise tracking for dynamic scenes; Geo4D [2] adapts video diffusion models. All require expensive supervision: DUST3R/VGGT use structure-from-motion labels, Geo4D uses synthetic data rendered from 3D assets, limiting real-world scalability.

2.2 Self-Supervised Learning (SSL)

Self-supervised learning (SSL) extracts training signals from data structure. Masked autoencoders [4] reconstruct masked patches; DINO [5, 10] matches predictions between augmented views, producing semantic features without labels. VideoMAE [11] extends masking to spatiotemporal tubes. For geometry, self-supervised depth [6, 12] uses photometric consistency between consecutive frames. However, methods predict per-frame depth independently, treating video as image sequences rather than modeling temporal geometric structure. Whether SSL objectives can teach spatiotemporal reasoning - tracking evolving geometry and maintaining multi-view consistency - remains unexplored yet fundamental to scaling video geometric understanding.

3 Problem Statement

While SSL has transformed image understanding, its application to video geometry remains limited by several fundamental challenges:

Appearance vs geometry - Geometric learning requires different signals than appearance-based pretraining. While methods like DINO learn semantic features through augmentation invariance, geometry demands precise metric structure and multi-view consistency. Models can minimize photometric loss through surface appearance matching without capturing underlying 3D structure. Do appearance-based objectives transfer to geometric reasoning, or is geometry-specific supervision necessary?

Temporal signals and shortcuts - Videos encode geometric constraints through motion parallax and occlusion patterns, yet naive temporal objectives permit trivial solutions - copying pixels across frames or learning optical flow without depth understanding. Which temporal SSL objectives force genuine spatiotemporal geometric reasoning rather than enabling such shortcuts?

Scaling to real video - Current methods use expensive supervision [1] or synthetic data [2], limiting scale. Whether pretraining on real unlabeled video can match these approaches, how much data is required, and which objectives transfer most effectively to downstream tasks remain uncharacterized - yet are essential for scaling video geometric learning.

4 Research Plan

4.1 Research Questions

(1) Which temporal SSL objectives effectively teach geometric structure? Beyond existing approaches like photometric consistency [6] or masked prediction [11], can we design novel objectives that leverage video-specific signals - cycle consistency across longer sequences, multi-view geometric coupling, or predictive temporal reasoning - to force genuine 3D understanding rather than surface appearance matching?

(2) How do different temporal signals interact in multi-objective training? SSL methods combine multiple objectives [5, 11] - temporal consistency, geometric coupling, predictive modeling. Which combinations are complementary versus redundant for geometric learning? Can we identify minimal sets that maximize geometric supervision while maintaining data efficiency?

(3) What are the data efficiency and scaling properties of video pretraining for geometry? How much unlabeled video is required to match supervised methods like [1] or synthetic approaches like [2]? Do scaling curves differ across objective types, and which downstream tasks benefit most from video pretraining?

4.2 Proposed Investigation

Methodological framework. This work systematically investigates SSL objectives across three categories: *temporal consistency* (photometric matching [6], cycle consistency), *geometric coupling* (depth-flow constraints, multi-view consistency [1]), and *predictive modeling* (masked spatiotemporal reconstruction [11]). Beyond evaluating existing methods, this research designs and tests novel objectives that exploit video-specific geometric signals. Through controlled experiments isolating individual design choices - objective type, combination of objectives within same type, and data scale - it establishes which approaches enable genuine geometric reasoning versus appearance-based shortcuts.

Experiment design. Pretraining uses real unlabeled video from RealEstate10K [13], containing diverse indoor and outdoor scenes with natural camera motion. Evaluation benchmarks include NYU Depth V2 [14] and KITTI [15] for depth estimation, Sintel [16] for optical flow, and standard metrics for camera pose estimation. Comparisons against supervised baselines [1] and synthetic pretraining approaches [2] measure both downstream task performance and data efficiency curves across varying amounts of pretraining data.

Timeline. Year 1 establishes the training framework, evaluates existing temporal SSL objectives, and explores novel objectives leveraging video-specific geometric signals. Year 2 conducts comprehensive ablation studies across all objective categories (temporal consistency, geometric coupling, predictive modeling) and investigates multi-objective interactions. Year 3 characterizes data efficiency through scaling experiments and validates transfer to downstream tasks (depth, pose, flow estimation).

4.3 Expected Contributions

This work establishes systematic understanding of which SSL objectives teach video geometry, how temporal signals interact in multi-objective training, and the data requirements for matching supervised methods. By identifying minimal objective combinations that force genuine geometric reasoning while avoiding shortcuts, this research provides foundational knowledge enabling scalable geometric learning from unlabeled video. Open-source implementations will support reproducibility and future research in data-efficient video geometric learning.

I would like to pursue this research under the supervision of Dr. Iro Laina and Dr. Christian Rupprecht, whose work on self-supervised learning from video [17, 18] and 3D geometric understanding [1] provides good guidance for this investigation.

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